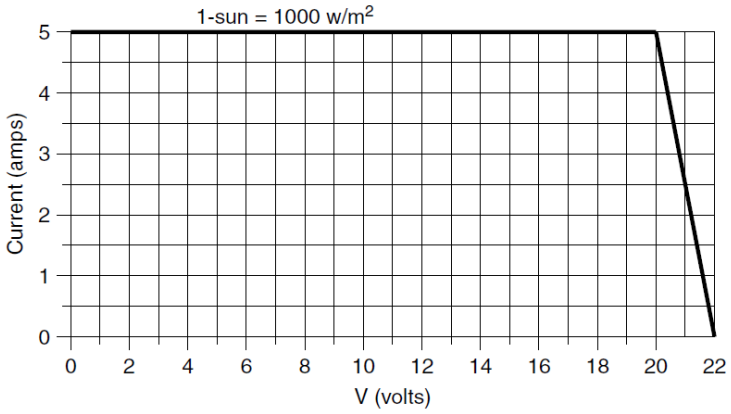
EEE407 Tutorial 6

1. A hydropower station has a gross head of 8.2 meter. Head loss in water conductor system is 0.5 meters. The average water flow rate is 267 cubic meter per second. With overall system's efficiency of 85%, what is the average power output and energy generated per year?

2. Suppose a 10cm-diameter penstock delivers 0.0095 m^3/s of water through an elevation change of 30m. The pressure in the pipe is 186k N/m^2 when it reaches the powerhouse. What fraction of the available head is lost in the pipe? What power is available for the turbine?

3. Suppose the I –V curve for a PV module exposed to 1-sun (1 kW/m^2) of insolation is as shown below:



   a. What load resistance R would result in the maximum power delivered to the panel at 1-sun? How much power would that be (watts)?
   b. Suppose the hour-by-hour insolation striking this single panel is given in the following table. For simplicity, consider each I –V curve to apply for one hour (e.g., the "8-A.M." curve is from 7:30 A.M. to 8:30 A.M. the insolation averages 200 W/m^2):

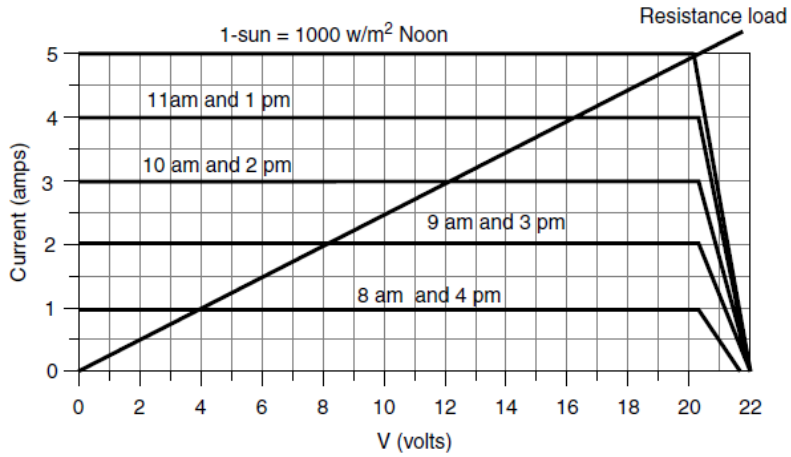
Time	Insolation (W/m^2)
8A.M., 4P.M.	200
9, 3	400
10, 2	600
11, 1	800
noon	1000

   If current (at any given voltage) is directly proportional to insolation, carefully draw the hour-by-hour I –V curves for the module. For the resistive load

determined in (a), what would be the energy (watt-hr) delivered by this panel in a day's time?

c. If the module is equipped with a maximum-power-point tracker, what energy would be delivered in a day's time? What percentage improvement does it provide?

4. Consider a single PV module supplying power to a resistance load. The hour-by-hour I –V curves for the PV module are shown below along with the I –V curve for the resistance. For simplicity, consider each I –V curve to apply for one hour (e.g., the "8-A.M." curve is from 7:30 A.M. to 8:30 A.M.).

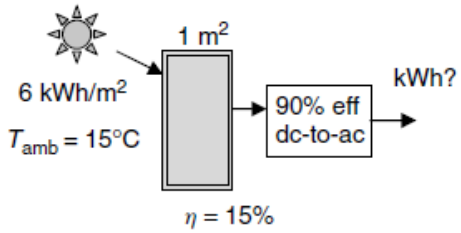


a. What is the resistance of the load?

b. Suppose the same resistive load is connected to two such modules wired together in series (so their voltages add). Draw the I –V curves for the pair of modules and from that, determine the daily energy delivered (W-hr). Compare that with the energy that would have been delivered with just a single module.

c. If two modules are connected in parallel (so their currents add), what would be the daily energy delivered to the same resistive load?

5. A clean, 15-%-efficient module (STC), 1 m^2 in size, has its own 90% efficient inverter. Its NOCT is 44°C and its rated power degrades by 0.5%/°C above the 25°C STC.



a. What would its STC rated power be?

b. Find the kWh that it would be expected to deliver on a 15°C Day with daylong insolation equal to 6 kWh/m^2?